

PROSPER PLATFORM

What the platform can do today

Comprehensive summary of every capability already shipped in the Prosper platform: Backoffice for internal staff, Client / Partner Portal, public Developer Portal, and the core infrastructure that powers tokenised-yield operations on Stellar.

1. At a glance

- **~50 backend endpoints** (FastAPI · MongoDB · async).
- **33 frontend screens** across three surfaces (Backoffice 19 + Portal 13 + Public 1).
- **49 / 49 automated tests** passing (auth, ops queue, KYC, webhooks, etc.).
- **Two-signer approvals · MFA TOTP · HMAC webhooks · idempotency · scoped multi-tenancy.**
- **Auto-seeded demo dataset** for an instant interactive walkthrough.
- **Brand-matched UI** with prosper.foundation (light + dark mode, IBM Plex / Chivo).

2. Backoffice (Prosper internal staff)

19 screens organised around a sidebar with environment switcher (Sandbox / Production) and surface switcher (Backoffice ↔ Portal).

Dashboard Aggregated KPIs (AUM, supply, NAV chart, 14-day volume), ops queue overview.	Clients List of partner organisations with filters by environment, type, status.	Client Detail Per-org view with contact, API keys, positions, transactions and document panel.
Onboarding List of KYC/KYB cases with SLA, progress, status filters.	Compliance Review queue: KYC / AML / sanctions / PEP / travel-rule decisioning.	Funds PROS fund management + NAV history + supply controls.
Products Term staking & liquid products with APR configuration.	Positions Every open / matured / redeemed position across the platform.	Treasury Issuer, treasury, reward pool and fee accounts.
Transactions Full ledger with filters, search by tx-hash, date-range and CSV export.	Reconciliation On-chain ↔ off-chain match queue with resolve workflow.	Integrations Partner API apps + real-time Prosper upstream status semaphore.
API Keys Issue / revoke API keys (plaintext shown once).	Webhooks Endpoint registry + delivery log + HMAC signing secret.	Alerts Operational alerts with severity + resolve action.

Reports NAV, audit, performance and tax reports.	Approvals Two-signer operations queue (mint, large redemption, etc.).	Audit Log Immutable trail of every action with filters by actor / resource / time.
Users & Roles Platform admin: list, role update, MFA status.		

3. Client / Partner Portal

13 screens scoped automatically to the partner's own organisation. Internal users can also switch into a portal view to support clients.

Overview My Portfolio: principal invested, accrued yield, active positions, NAV chart.	Organization Corporate info + KYC / KYB document upload panel.	Users Invite / list / manage org members and their roles.
Balances USDC + PROS balances per environment.	Transactions Org-scoped ledger with filters and CSV export.	Yield & Performance APR by product + historical yield paid out.
End Customers Sub-investors managed under the partner umbrella.	Integrations Configure partner API apps and connected services.	API Keys Self-service: issue and revoke keys for own org.
Webhooks Register endpoints + view HMAC signing secret + delivery log.	Reports NAV, tax and performance reports filtered to the org.	Compliance Track the org's own compliance status and history.
Settings Account preferences, MFA, session management.		

4. Public Developer Portal

Unauthenticated route `/developers` — the front door for technical integrators.

- **Hero + Quickstart** — get an API key, call subscribe, listen to webhooks (30-second walkthrough).
- **Endpoint reference** with sidebar navigation: Auth, Subscribe & Redeem, Mint (two-signer), Webhooks, Transactions & Audit.
- **Live language tabs** — every endpoint has executable curl / JavaScript / Python examples with copy-to-clipboard.
- **OpenAPI 3.1** machine-readable schema served at `/openapi.json`.
- **Webhook signature recipe** — HMAC-SHA256 with timestamp, ready-to-paste verification snippets.

5. Core business flows

SUBSCRIBE	Partner posts POST <code>/api/positions/subscribe</code> with amount + product. Backend creates a position, a transaction, a reconciliation stub, and proxies to the Stellar rail. <code>prosperTxId</code> (UUIDv4) anchors idempotency end-to-end.
REDEEM	POST <code>/api/positions/{id}/redeem</code> settles principal + accrued interest. Ownership check + audit log + reconciliation entry.
MINT (two-signer)	POST <code>/api/transactions/mint</code> never executes directly — it creates an <code>apv_*</code> approval that a different internal user must approve. MFA challenge enforced if the approver has TOTP enabled.

APPROVALS	GET /api/approvals/pending + POST /api/approvals/{id}/approve — full operations queue with signatures and audit trail.
RECONCILIATION	Background worker matches on-chain transactions against the off-chain ledger by prosperTxId; failures land in a review queue with manual resolve.
WEBHOOKS	Register endpoints, receive signed POSTs with X-Prosper-Signature: t=..., v1=.... Automatic retries with exponential backoff; full delivery log.
DOCUMENTS	Upload KYC / KYB files (PDF, PNG, JPG, WebP, CSV — up to 10 MB) with magic-byte validation, soft-delete and per-org scoping. Stored in Emergent managed object storage.
GLOBAL SEARCH (■K)	Single endpoint /api/search?q= indexes organisations, transactions, positions, API keys and onboarding cases. Wired into the ■K command palette in the UI.

6. Security & compliance features

MULTI-TENANT SCOPING	Every read and write is filtered by <code>org_id</code> for non-internal users. <code>@prosper.foundation</code> emails are auto-flagged super-admin.
EMERGENT GOOGLE OAUTH	Sessions issued by Emergent's managed OAuth service. 7-day expiry, <code>httpOnly</code> cookie with <code>samesite=none</code> ; also accepts <code>Authorization: Bearer</code> for API clients.
MFA TOTP	Per-user TOTP setup / verify / disable via <code>/api/auth/mfa/*</code> . Challenge enforced at approval time for high-value operations.
HMAC WEBHOOK SIGNING	Every outbound webhook is signed with the endpoint's secret using HMAC-SHA256. Plaintext secret is returned ONCE at endpoint creation.
IDEMPOTENCY	All write mutations accept an <code>Idempotency-Key</code> header; <code>prosperTxId</code> serves as the reconciliation anchor.
AUDIT LOG	Every privileged action is recorded with actor, resource, metadata and timestamp. Queryable by filters; immutable by design.
MAGIC-BYTE VALIDATION	Document uploads validate the actual file signature (PDF, PNG, JPEG, WebP) and UTF-8 heuristic for CSV. Renaming a binary to <code>.pdf</code> is rejected with HTTP 400.
CROSS-ORG GUARDS	Non-internal users cannot read or write to another org — verified by automated test suite (<code>test_client_admin_cannot_upload_to_other_org</code>).

7. Operational features

- **■K command palette** with global search and quick navigation across the entire UI.
- **Alerts bell** in the top bar with unresolved counter and click-through.
- **Environment switcher** (Sandbox ↔ Production) propagating to every data query.
- **Surface switcher** (Backoffice ↔ Portal) for internal users supporting clients.
- **Background workers** (APScheduler): webhook retries, SLA checks, end-of-day NAV snapshots.
- **Real-time upstream status semaphore** on Integrations: enabled / reachable / authenticated / error.
- **CSV exports** on transactions, positions, reconciliation and reports.
- **Date-range filters** with pagination on every large list.
- **Light + Dark theme** with persistent preference, matched to `prosper.foundation`.
- **Demo banner** on every page when `is_demo` records exist.
- **Quick-demo entry points** from the landing page → instant Backoffice or Portal access without OAuth (seeded `super_admin` / `client_admin` tokens).

8. Developer & integration capabilities

- **OpenAPI 3.1** auto-generated from FastAPI; served live at `/openapi.json`.
- **API keys** with environment scoping (sandbox / production); plaintext shown once; revocable.
- **Webhook registry** with selectable event types and HMAC signing secret per endpoint.

- **Stable session tokens** for automation — seeded on every backend boot.
- **Prosper upstream proxy** (`prosper_client.py`) — ready to switch from simulated to real Stellar calls with one `.env` flag.
- **Diagnostic endpoint** `GET /api/admin/prosper-upstream` reports connectivity health.
- **Modular routers** — one file per domain under `/app/backend/routers/` for maintainability.

9. Test coverage & quality

- **49 / 49 pytest cases passing** covering auth, dashboard, organisations, onboarding, compliance, funds, products, positions, treasury, transactions, reconciliation, integrations, alerts, reports, audit, users.
- **Documents API:** 14 dedicated tests (upload happy path, list, download, soft-delete, magic-byte spoof rejection, .exe rejection, .zip rejection, cross-org 403 guard, client own-org happy path).
- **Admin diagnostics:** 3 tests for the Prosper upstream endpoint (shape, role guard, unauth rejection).
- **Four iterations** of the testing agent confirmed end-to-end flows.
- **Modular architecture** — 19 router files + shared helpers; storage.py on async httpx so the event loop is never blocked by 10 MB uploads.

10. What is still simulated

PROSPER STELLAR RAIL

Calls to the real Prosper Stellar API are **code-ready** but currently mocked because the pod's network egress to lb-backend-develop-*.elb.amazonaws.com is closed. Once infra opens it, set `PROSPER_API_ENABLED=true` in backend/.env and the Integrations status card turns green automatically. Affected: subscribe, redeem, mint.

Demo data (5 orgs, 2 funds, 5 products, 35 positions, 120 transactions, 120 reconciliations, 24 end customers, etc.) is tagged `is_demo: true` and can be removed with a single call to `POST /api/admin/wipe-demo` without affecting real data.

11. Pre-seeded for instant exploration

On first boot the backend auto-seeds a complete demo environment so the platform is fully interactive out of the box. Tokens are refreshed every backend restart.

Organisations	5 (Alemany Capital, Beachside Wealth, Quirón Markets, Meridian Sandbox, Andes Trust)
Funds + products	1 production fund (PROS / Quirón PyMEs) + 1 sandbox + 5 products
Positions + transactions	35 positions · 120 transactions · 120 reconciliation records
API surface seed	6 apps · 3 keys · 3 webhooks · 18 deliveries
Compliance & ops	4 alerts · 12 reports · 40 audit events · 24 end customers
Test users	super_admin (token test_session_prosper_super_admin) + client_admin (token test_session_prosper_client_admin) — both refreshed on every boot.

BOTTOM LINE

The platform is demo-ready, end-to-end tested and deployable. The whole control plane — auth, multi-tenancy, MFA, approvals, webhooks, idempotency, audit, documents, ops queue — is real. The single remaining hop to a productive first client is the Prosper Stellar upstream egress (infra task, no code).